

- 19** Light of wavelength λ is incident normally on a diffraction grating of slit spacing five times the wavelength of the light.

What is the angle of diffraction of the second order maximum?

A	12°	B	24°	C	37°	D	45°
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Answer: B

$$\begin{aligned}
 d \sin \theta &= n \lambda \\
 \sin \theta &= \frac{n \lambda}{d} \\
 &= \frac{2 \lambda}{5 \lambda} \\
 &= \frac{2}{5} \\
 \theta &= 24^\circ
 \end{aligned}$$

- 20** The figure below shows two horizontal parallel metal plates. The upper plate is maintained at